C3.2 Defence against disease

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c3-2.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Image credits are on the last slide of the slides for this subtopic.

GETTING IN

C3.2.1 Pathogens are what cause infectious disease

pp. 517–518

A **pathogen** is anything living or virus-like that causes disease in a **host**. Because pathogens can pass between hosts, the illnesses they cause are called **infectious** or communicable. That separates them from conditions such as cancer, malnutrition or inherited disorders, which you cannot catch.

The word is mainly used for four groups, and each has features you can use to identify it:

- **Bacteria:** prokaryotes with a peptidoglycan wall. Most species are harmless; those that are not include the causes of cholera and tuberculosis.
- **Viruses:** not cells. A virus is nucleic acid in a protein coat and can reproduce only inside a host cell, so every virus is a parasite.
- **Fungi:** eukaryotes with chitin walls, responsible for infections such as athlete's foot.
- **Protists:** single-celled eukaryotes, including the parasite that causes malaria.

Some parasitic worms also cause human disease. No archaeon is known to be a human pathogen, and scientists still debate why.

C3.2.2 How skin and mucous membranes act as the first defence

pp. 519–520

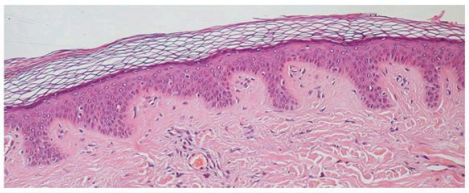
The first job is to stop pathogens entering at all.

Skin defends in two ways. As a **physical** barrier, its outer layer of dead, keratin-filled cells is tough and sealed with lipid, and shedding removes microbes that settle on it. As a **chemical** barrier, its secretions lower the surface pH and sweat contains lysozyme, an enzyme that damages bacterial walls.

Mucous membranes line the airways, gut and other openings. They are thin and living, so they need more protection. Mucus traps particles, and in the airways, **cilia** sweep it up to the throat to be swallowed. Stomach acid then kills many of the microbes that arrive.

Skin and mucous membranes

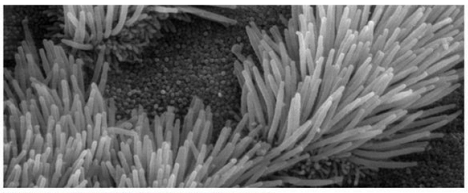
C3.2.2



Skin: the pale top layer is dead, keratin-filled cells. Kilbad, public domain.

Physical: tough outer layer of dead cells packed with **keratin**, sealed by lipid.

Chemical: sebum and sweat keep the surface acidic, and **lysozyme** damages bacterial walls.



Cilia lining an airway (SEM). Charles Daghlian, public domain.

Line the airways, gut and other openings: thin and living, so easier to cross.

Sticky **mucus** traps microbes; **cilia** sweep it up to the throat to be swallowed. Stomach acid kills many.

Think
Why do smokers, whose cilia are damaged, get more chest infections?

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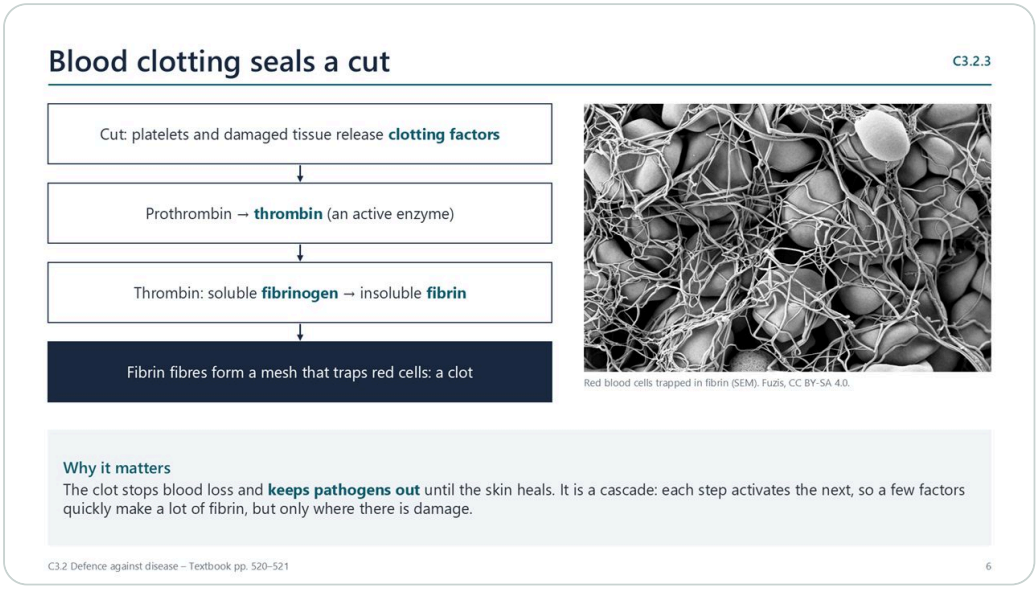
Exam note: you do not need to draw or label skin.

C3.2.3 How blood clotting closes cuts in the skin

pp. 520–521

A cut breaks the barrier, so it must be sealed quickly. Doing this limits blood loss and keeps pathogens out.

1. **Platelets** collect at the wound and stick together, which is enough to plug a tiny break.
2. Platelets and damaged tissue release **clotting factors**, starting a cascade.
3. The cascade converts prothrombin into the enzyme **thrombin**.
4. Thrombin rapidly converts soluble **fibrinogen** into insoluble **fibrin** fibres.
5. The fibres form a mesh that traps **erythrocytes**, and this is the clot.



Watch out: fibrinogen is the soluble starting material and fibrin is the insoluble product. Write that a clot "prevents the entry of pathogens", not that it "stops diseases getting in".

FIGHTING BACK

C3.2.4 How innate and adaptive immunity differ

p. 522

	INNATE	ADAPTIVE
RESPONDS TO	Broad categories of pathogen	One particular pathogen
DURING YOUR LIFE	Stays the same	Builds a memory of pathogens met
ON RE-EXPOSURE	No improvement	Faster and more effective
CELLS TO KNOW	Phagocytes	Lymphocytes

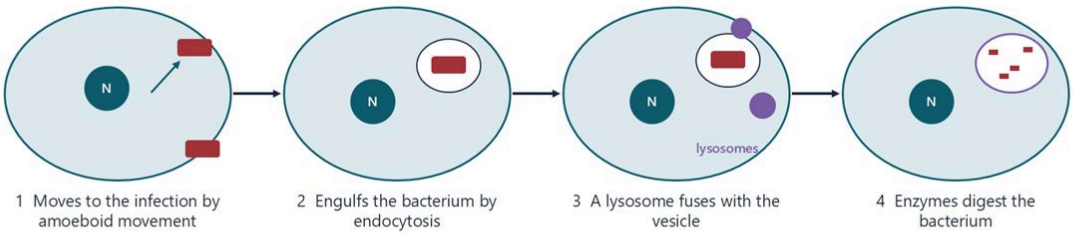
Both start when a pathogen gets past the surface barriers. The only innate component you must know is the phagocyte.

C3.2.5 How phagocytes help to control infection

pp. 522–523

Phagocytes engulf and digest

C3.2.5

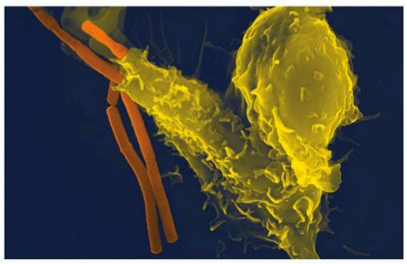


1 Moves to the infection by amoeboid movement

2 Engulfs the bacterium by endocytosis

3 A lysosome fuses with the vesicle

4 Enzymes digest the bacterium



Neutrophil (yellow) engulfing bacteria (orange), coloured SEM. Volker Brinkmann, CC BY 2.5.

Phagocytes squeeze out of capillaries and move to the infected tissue. They recognise pathogens as foreign, take them in, and digest them with **lysosome** enzymes.

They attack **any** pathogen of a broad type, so they are part of the innate system.

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Phagocytes are white blood cells that leave the blood by **amoeboid movement**, squeezing through capillary walls and crawling to the infected tissue. They recognise a pathogen, engulf it by **endocytosis** so it sits inside a vesicle, then fuse a **lysosome** with that vesicle. Lysosomal enzymes digest the pathogen.

C3.2.6 How lymphocytes work together in adaptive immunity to make antibodies

pp. 523–524

Lymphocytes circulate in the blood and are also held in **lymph nodes**. Two kinds matter here: **B-cells**, which go on to make antibodies, and **helper T-cells**, which activate B-cells.

You carry a huge number of different B-cell types. Each type can make one specific antibody. That variety lets the body respond to almost any antigen, but it also means only a few cells match any one antigen when it first appears.

C3.2.7 Antigens: molecules that are recognised and set off antibody production

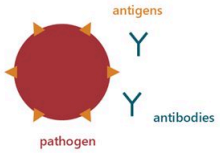
An **antigen** is a molecule that the immune system recognises as foreign and that triggers antibody production. Most are glycoproteins or other proteins, usually on the pathogen's outer surface.

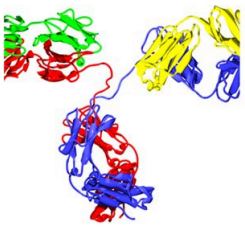
An **antibody** is a Y-shaped protein made by B-cells. The binding site at the tip of each arm fits one particular antigen.

Red blood cells have antigens too. Group A cells given to someone with group B blood will be recognised as foreign, which can stimulate antibody production. This is why donated blood is matched.

Antigens and antibodies

C3.2.7





An antibody, about 10 nm tall (PDB 1IGT). TimVickers, public domain.

An **antigen** is a molecule recognised as foreign, usually a protein or glycoprotein on the pathogen surface.

An **antibody** is a Y-shaped protein. The tips of its arms fit one antigen exactly.

Blood group	Antigens on red cells	Antibodies in plasma
A	A	Anti-B
B	B	Anti-A
O	None	Anti-A and anti-B

Blood groups

Red cell antigens trigger antibodies too. Giving the wrong group makes the donor cells clump.

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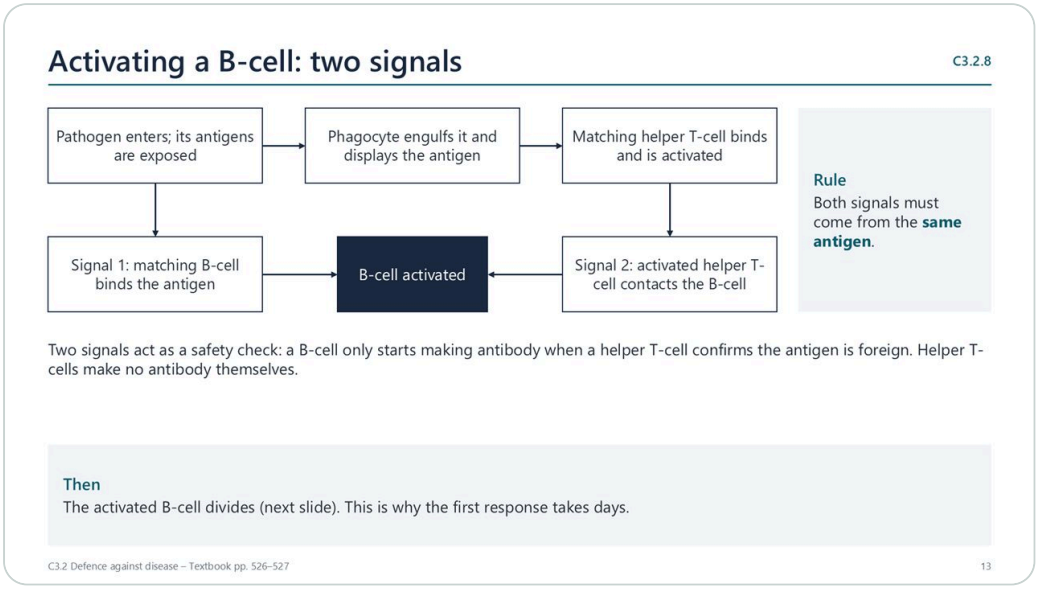
C3.2.8 How helper T-cells activate B-cells

pp. 526–527

A B-cell stays inactive until it receives **two signals**:

1. Direct contact with its specific antigen.
2. Contact with a helper T-cell that has been activated by the same antigen.

A phagocyte that has engulfed a pathogen displays pieces of it on its surface. A helper T-cell with the matching receptor binds to these and becomes activated. It then finds and activates the B-cell that matches the same antigen.



Watch out: helper T-cells do not secrete antibodies. Their role is to activate the B-cell.

C3.2.9 How activated B-cells multiply into clones of plasma cells that secrete antibody

pp. 527–528

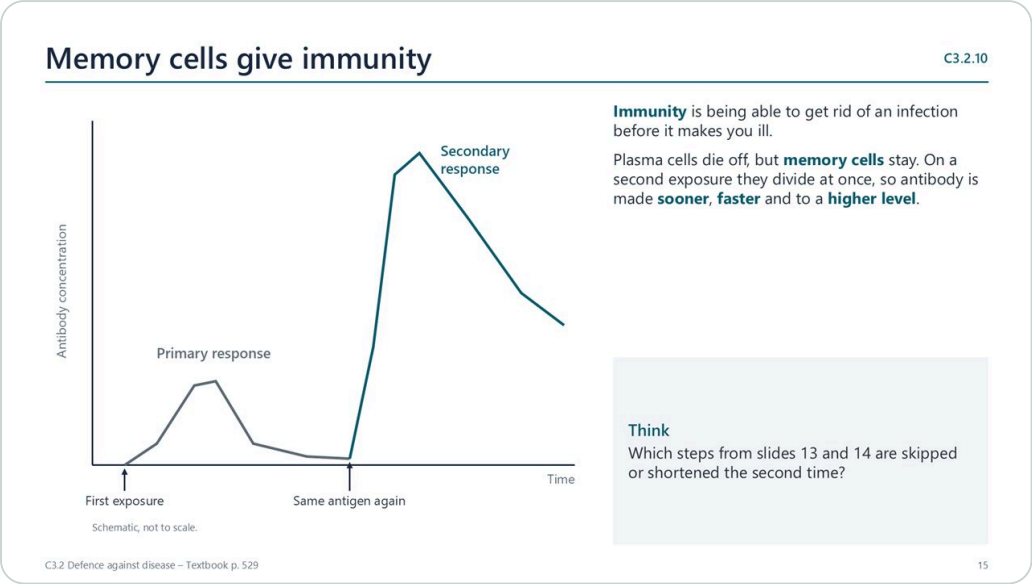
Too few B-cells match any one antigen to make enough antibody. So an activated B-cell first divides repeatedly by **mitosis**, forming a clone of identical cells. Most of them become **plasma cells**, whose rough ER makes large amounts of the same antibody for secretion. This process is known as clonal selection.

C3.2.10 How keeping memory cells gives immunity

p. 529

Immunity means being able to eliminate an infectious disease from the body. After an infection, plasma cells and antibodies decline, but some activated lymphocytes survive for the long term as **memory cells**.

If the same antigen returns, the memory cells respond at once. Antibodies are produced **more rapidly and to a higher level**, often before you feel ill. Use both comparisons in an exam answer.



WHEN DEFENCE FAILS

C3.2.11 How HIV passes between people in body fluids

p. 530

HIV passes from one person to another in body fluids, for example through sexual intercourse, shared hypodermic needles, breastfeeding and transfusions or transplants from unscreened donors. It does not pass through sharing a glass or towel, and mosquitoes do not transmit it.

The implications are practical. Condoms, clean needles and screening of donated blood all break the route of transmission, and knowing how HIV does not spread helps to reduce stigma.

Watch out: HIV is transmitted. AIDS is the condition that can follow, and it is not itself transmitted.

C3.2.12 How HIV infects lymphocytes and leads to AIDS

pp. 529–531

HIV is a retrovirus. Proteins on the virus bind to receptors called CD4 on helper T-cells, and its RNA core enters the cell. The enzyme reverse transcriptase copies the RNA into DNA, which is inserted into a host chromosome. The cell can then make new viruses, and over time infected lymphocytes are lost.

Only certain types of lymphocyte are infected, but losing them has wide effects. With fewer helper T-cells, fewer B-cells are activated and antibody production falls. The body then cannot fight **opportunistic infections** that a healthy person would normally overcome. This stage is **AIDS**, acquired immunodeficiency syndrome.

C3.2.13 Antibiotics: chemicals that block processes in bacteria but leave eukaryotic cells unharmed

pp. 532–534

Antibiotics can treat a patient safely because bacteria and human cells differ. They target features found only in prokaryotes, such as cell wall synthesis or 70S ribosomes. Human cells have no cell wall and have different ribosomes in their cytoplasm, so they are not harmed.

Viruses have no cell wall, no ribosomes and no metabolism of their own. There is nothing for an antibiotic to act on, so it will not control a viral infection.

Antibiotics: bacteria only

C3.2.13

Process blocked	In bacteria	In human cells
Cell wall synthesis	Peptidoglycan wall weakens; cell bursts	No cell wall
Protein synthesis	70S ribosomes blocked	80S ribosomes unaffected
Enzymes for DNA or metabolism	Bacterial enzymes blocked	Different enzymes



Penicillium, source of penicillin. Crulina 98, CC BY-SA 3.0.

Why safe
Antibiotics hit structures that **prokaryotes** have and our cells do not, so they kill bacteria inside a patient without harming the patient.

Why useless on viruses
Viruses have **no wall, no ribosomes and no metabolism** of their own: they use the host cell. Antivirals target viral steps, such as reverse transcriptase.

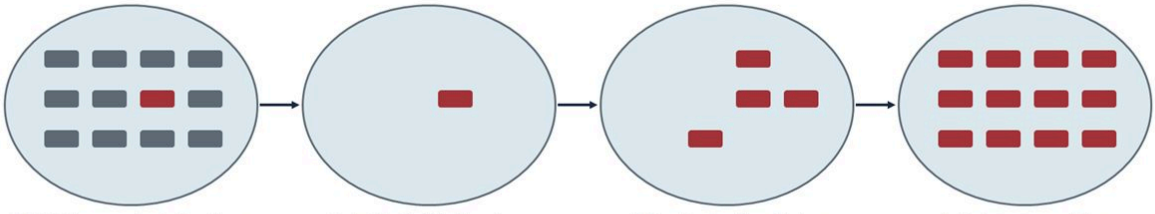
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Watch out: it is wrong to say that antibiotics fail because viruses "hide inside cells". The reason is that viruses lack the structures antibiotics act on.

C3.2.14 How pathogenic bacteria evolve resistance to several antibiotics

Resistance evolves by natural selection

C3.2.14



1 Variation: one bacterium has a resistance mutation

2 Antibiotic kills the others

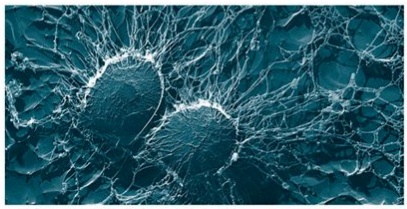
3 Resistant cell multiplies

4 Resistant population

red = resistant; grey = killed by the antibiotic

The antibiotic does not cause the mutation: it **selects** cells that already have it.

Resistance genes are often on **plasmids**, which can pass between bacteria, so one strain can collect resistance to several antibiotics.



Staphylococcus aureus, which includes MRSA (SEM). USDA, public domain.

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This is natural selection. A random mutation, or a gene gained from another bacterium on a plasmid, makes a cell resistant. When an antibiotic is used, the susceptible bacteria die and the resistant ones survive and reproduce with less competition. Repeating this with different antibiotics can produce **multiresistant** strains such as MRSA.

Careful use slows this down: prescribing antibiotics only when needed, finishing the course, keeping hospitals hygienic and limiting routine use in farming. It matters because new antibiotics are hard to develop.

CROSSING SPECIES

C3.2.15 Zoonoses: infectious diseases that can pass from other species to humans

pp. 537–540

A **zoonosis** is an infectious disease that can transfer from another species to humans. Many pathogens stick to one host, but others cross, especially when people and animals live closely together.

DISEASE	PATHOGEN	ANIMAL SOURCE	ROUTE TO HUMANS
TUBERCULOSIS	Bacterium (zoonotic form: *M. bovis*)	Cattle	Raw milk, contact with infected animals or tissue
RABIES	Virus	Dogs and other mammals	Bite, or saliva on broken skin
JAPANESE ENCEPHALITIS	Virus	Pigs and wading birds	Mosquito bite
COVID-19	Virus (SARS-CoV-2)	Probably from bats; exact route not confirmed	Now spreads between humans

COVID-19 shows how serious a recent transfer can be, with worldwide effects on health and daily life.

PROTECTING EVERYONE

C3.2.16 How vaccines and immunization work

pp. 540–542

A vaccine contains **antigens**, or **DNA or RNA** coding for antigens. It may be a weakened or inactivated pathogen, a purified part of one, or a nucleic acid such as the mRNA in some COVID-19 vaccines. It stimulates a primary immune response and the production of memory cells, without causing the disease. If the real pathogen arrives later, the fast secondary response deals with it.

▶ **Going further: active and passive immunity**

C3.2.17 How herd immunity helps to prevent epidemics

pp. 544–545

A population does not need everyone to be immune to stop a disease. When a high enough percentage is immune, the pathogen seldom reaches a susceptible person, so transmission is greatly impeded. This is **herd immunity**.

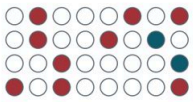
Its members depend on each other: those who can be vaccinated help protect those who cannot. The percentage needed varies with the disease and is higher for very contagious ones such as measles.

Herd immunity

C3.2.17



Few immune: infection spreads



Most immune: infection stalls



Not immune Infected Immune

Very contagious diseases need more people immune: about 95% for measles.

Brooklyn Bridge walkway. EazyDee107, CC0.

Key idea

When most people are immune, an infected person rarely meets anyone who can catch it, so the chain of transmission breaks. This also protects people who cannot be vaccinated, such as newborns and people with weak immune systems.

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C3.2.18 Evaluating data from the COVID-19 pandemic

pp. 546–547

Percentage change

(new – original) ÷ original × 100

Over time, or vaccine compared with placebo. In one vaccine trial, 162 cases with placebo and 8 with vaccine, in groups of similar size: (162 – 8) ÷ 162 × 100 = 95.1% fewer cases.

Percentage difference

difference ÷ average × 100

Two values at the same time. Invented example: 150 and 90 cases per 100 000 in two countries: (150 – 90) ÷ 120 × 100 = 50%.

When you evaluate data, think about what else could explain a pattern, such as the amount of testing, how cases are reported, or differences in public health rules. Try it with real figures on the WHO COVID-19 dashboard; slide 27 gives the steps.

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